METRIC

MILITARY SPECIFICATION

PIN, SPRING, TUBULAR, (COILED AND SLOTTED), GENERAL SPECIFICATION FOR, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tubular-shaped spring pins constructed to resist reduction in pin diameter by means of spring action.

1.2 Classification. Spring pins shall be of the following types and classes as specified on the applicable specification sheets.

1.2.1 Types. Spring pins shall be the following types:

Type I - Slotted
Type II - Coiled

1.2.2 Classes. Type II spring pins shall be of the following classes:

Class 1 - Heavy duty
Class 2 - Standard duty
Class 3 - Light duty

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements for
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging and Packing for Shipment and Storage of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DOD-P-16232 - Phosphate Coating, Heavy, Manganese or Zinc base (For Ferrous Metals).
MIL-C-13924 - Coating, Oxide, Black for Ferrous Metals.
MIL-I-17214 - Indicator, Permeability; Low-MU (Go-No Go).

(See supplement 1 for list of associated specification sheets.)

STANDARDS

FEDERAL

FED-STD-66 - Steel: Chemical Composition and Hardenability.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1312 - Fasteners, Test Methods.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ANSI/ASTM A380-78 - Standard Recommended Practice for Cleaning and Descaling Stainless Steel Parts, Equipment and Systems.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and specification sheet, the latter shall govern.

3.2 Materials. Recycled and reclaimed materials (see 3.1) shall be used to the maximum extent practicable.

3.2.1 Carbon steel. Carbon steel shall conform to any of the steel compositions Nos. 1070 thru 1095, as specified in FED-STD-66.

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3.2.2 Corrosion-resistant steel (austenitic). Austenitic corrosion-resistant steel shall conform to the composition of steel number 302 as specified in FED-STD-66.

3.2.3 Corrosion-resistant steel (martensitic). Martensitic corrosion-resistant steel shall conform to the composition of steel no. 420, as specified in FED-STD-66.

3.3 Dimensions. Dimensions and tolerances for all types of spring pins shall conform to the dimensional requirements of the applicable specification sheet and shall apply after protective finishes.

3.4 Performance characteristics. The spring pins shall meet the following performance requirements when subjected to the applicable test specified in 4.8.

3.4.1 Magnetic permeability. Magnetic permeability of the spring pins made of 302 austenitic corrosion-resistant steel shall not be greater than 2.0 (Air = 1.257 μH/m) (microHenry per meter) at a field of 16 kA/m (kiloampere per meter) (see 4.8.4).

3.4.2 Double shear strength. The double shear strength of the spring pins shall be as specified on the applicable specification sheet. The spring pins shall be capable of withstanding the minimum double shear strength test for the respective materials (see 4.8.5). Spring pins which have been tested for double shear strength shall show a ductile shear without longitudinal cracks (see 4.8.6).

3.5 Finish. The protective finish of the spring pins when specified in the applicable specification sheet shall be as follows (see 4.8.7).

3.5.1 Black oiled finish. Carbon steel spring pins shall be furnished with a black, tumbled, oil dipped finish in accordance with MIL-C-13924, class 1.

3.5.2 Cadmium plate. Carbon steel spring pins shall be cadmium plated in accordance with QQ-P-416, type II, class 2. Plating thickness shall be 7.6 μm (micrometers).

3.5.3 Passivate. Corrosion-resistant steel spring pins shall be passivated in accordance with ASTM A380.

3.5.4 Phosphate coating. Carbon steel spring pins shall be phosphate coated in accordance with DOD-P-16232, type Z, class 2. Phosphate coating shall be 5 to 15 μm (micrometers) thick.

3.5.5 Zinc coating. Carbon steel spring pins shall be zinc coated in accordance with QQ-Z-325, type I, class 3. Coating thickness shall be 5.1 μm (micrometers).

3.5.6 Hydrogen embrittlement relief. Embrittlement relief of spring pins shall be in accordance with the applicable finish specification specified in 3.5 herein.

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3.6 Hardness.

3.6.1 Carbon steel spring pins. Carbon steel spring pins shall have a hardness of 46HRC to 53HRC or equivalent.

3.6.2 Corrosion-resistant steel spring pins.

3.6.2.1 Type I and type II spring pins. Type I and type II spring pins made of 302 austenitic corrosion-resistant steel shall be hardened by cold working.

3.6.2.2 Type I slotted spring pins. Type I slotted spring pins made of 420 corrosion-resistant steel shall have a hardness of 43HRC thru 52HRC.

3.6.2.3 Type II coiled spring pins. Type II coiled spring pins made of 420 corrosion-resistant steel shall have a hardness of 46HRC thru 55HRC.

3.7 Straightness. Straightness of spring pins shall be within the limits necessary to permit spring pins to fall through a ring gage by their own weight (see 4.8.3).

3.8 Workmanship. The workmanship shall be uniform in quality and free of irregularities or detrimental defects. The surfaces shall not contain any foreign matter. Edges and corners, as applicable shall be broken or rounded.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Material inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.8.1 thru 4.8.9.

4.4 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials used in fabricating the spring pins are in accordance with the applicable paragraphs listed in 3.2.1 thru 3.2.3.

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4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in table I. Each spring pin which does not pass any of the tests specified shall be defective.

4.6 Quality conformance inspection sampling.

4.6.1 Inspection lot. An inspection lot shall consist of all spring pins covered by a single specification sheet produced under essentially the same conditions, and offered for inspection at any one time.

TABLE I. Quality conformance inspection.

Inspection	Requirement paragraph	Test method paragraph
Dimensions	3.3, 3.7	4.8.1, 4.8.2, 4.8.3
Magnetic permeability	3.4.1	4.8.4
Double shear strength	3.4.2	4.8.5, 4.8.6
Finish	3.5.1, 3.5.2, 3.5.3, 3.5.4, 3.5.5	4.8.7
Hydrogen embrittlement relief	3.5.6	4.8.8
Hardness	3.6	4.8.9.1, 4.8.9.2

4.6.2 Sampling for visual and dimensions examination. A random sample of spring pins shall be taken from each lot in accordance with MIL-STD-105 inspection level II. The acceptable quality level (AQL) shall be as indicated in table II. The AQL shall apply to each individual defect, not to a group of defects.

4.6.3 Sampling for magnetic permeability test. Sampling for magnetic permeability of corrosion-resistant steel spring pins shall be in accordance with MIL-STD-105 inspection level S-4, with an AQL of 2.5 percent defective.

4.6.4 Sampling for shear test. Sampling for shear test of steel spring pins shall be in accordance with level S-4 of MIL-STD-105. The AQL shall be 2.5 percent defective.

4.6.5 Sampling for finish test. Sampling for protective finish test of spring pins shall be in accordance with the applicable specification referenced in 3.5.1 thru 3.5.5.

4.6.6 Sampling for hardness test. Sampling for hardness test of steel spring pins shall be in accordance with level S-4 of MIL-STD-105. The AQL shall be 2.5 percent defective.

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4.6.7 Sampling for hydrogen embrittlement relief test. Sampling for hydrogen embrittlement relief test shall be in accordance with level S-4 of MIL-STD-105. The AQL shall be 2.5 percent defective.

TABLE II. Classification of defects.

Category	Defect	Inspection method
Critical	None defined	
Major 101 102 103	AQL = 1.5 percent defective Diameter Length Straightness	 SIE 1/ SIE SIE
Minor 201 202	AQL = 2.5 percent defective End chamfers Edges and corners broken	 SIE Visual

1/ Standard inspection equipment.

4.7 Packaging inspection. The sampling and inspection of the preservation, packaging, packing and container marking shall be in accordance with the requirements of PPP-H-1581.

4.8 Methods of inspection.

4.8.1 Visual and dimensional examination. Each spring pin taken as specified in 4.6.2 shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with table II.

4.8.2 Diameters.

4.8.2.1 Type I, slotted spring pins. The maximum diameter of the slotted spring pin as specified on the applicable specification sheet, shall be inspected with a "go" ring gage. This ring size shall have a length of hole not greater than 3.0 mm. The minimum diameter shall be determined by averaging the three diameters A-1, A-2 and A-3 shown on the applicable documents. These diameters shall be measured at center of the spring pin with lengths 25.0 mm or shorter. If the length of the pin is greater than 25.0 mm, the minimum diameter shall be measured 6.0 mm from each end.

4.8.2.1.1 Slot dimensions. The dimensions and end configurations of the slot in the circumference of slotted type spring pins shall be such that spring pins in the free state will not interlock and that the sides of slot will not contact when spring pin is inserted into a hole of a diameter equal to the recommended minimum hole size within a tolerance of ± 0.008 mm. The edges of slot intersecting the spring pin diameter shall be broken or rounded.

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4.8.2.2 Type II, coiled pins. The maximum diameter of the coiled spring pin, as specified on the applicable specification sheet, shall be inspected with a "go" ring gage. This ring gage shall have a length of hole not greater than 3.0 mm. The minimum diameter shall be inspected with a "no go" ring gage.

4.8.3 Straightness. The straightness of the spring pins shall be determined by using a ring gage. The maximum inside diameter of the gage shall be greater than the maximum diameter specified for the spring pins on the applicable specification sheet by the amount of the straightness tolerance. The spring pins shall fall through this gage by their own weight. The gages shall be as specified in tables III and IV.

TABLE III. Straightness limits for slotted spring pins (mm).

Pin length	Gage length		Gage hole diameter	
	max	min	min	max
			Specified pin dia max plus	Specified pin dia max plus
To and incl 25	25.15	24.85	0.20	0.21
Over 25 to 50	50.15	49.85	0.40	0.42
Over 50	75.15	74.85	0.60	0.63

TABLE IV. Straightness limits for coiled spring pins (mm).

Pin length	Gage length		Gage hole diameter	
	max	min	min	max
			Specified pin dia max plus	Specified pin dia max plus
To and incl. 24	25.15	24.85	0.19	0.20
Over 24 to 50	50.15	49.85	0.32	0.34
Over 50	75.15	74.85	0.45	0.48

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4.8.4 Magnetic permeability test. When specified (see 6.2), corrosion-resistant steel spring pins taken in accordance with 4.6.3 shall be tested for compliance to the magnetic permeability requirements of 3.4.1. The test indicator shall be in accordance with MIL-I-17214.

4.8.5 Double shear strength test. The double shear strength test shall be conducted as follows:

a. The shear plane shall be at least one spring pin diameter away from the ends of the spring pin.
b. For type I, slotted spring pins, the slot shall be located approximately 90 degrees to the line of application of force.
c. Spring pins too short to be tested in double shear shall be tested by shearing two spring pins simultaneously in a single shear.
d. The maximum clearance between the loading member and the supporting member shall be 0.15 mm. The shearing surfaces of the test fixtures shall have a minimum hardness of Rockwell C-58, or equivalent. The test plates shall have a minimum total thickness of two times the outside diameter of the spring pin.
e. The shear values specified on the applicable specification sheet shall apply to spring pins tested in the minimum recommended hole sizes that are specified herein.
f. Spring pin shall be tested to destruction to determine conformance to the specified shear values.

4.8.6 Ductility. Spring pins, subjected to the double shear strength test, shall be examined to determine conformance to the ductility requirement of 3.4.2.

4.8.7 Finish test. The carbon steel spring pins shall conform to the finish test of QQ-P-416, MIL-C-13924, DOD-P-16232 and QQ-Z-325 as specified in 3.5. Corrosion-resistant steel spring pins shall conform to finish test of ASTM A380.

4.8.8 Hydrogen embrittlement relief. The spring pins shall be tested for hydrogen embrittlement relief by the following method:

Plated or coated spring pins shall be installed in the minimum recommended hole size specified on the applicable specification sheet. The holes shall have a length of at least 75 percent of the length of the spring pin tested. The spring pins shall remain installed in these holes for a minimum of 100 hours without failure. Upon removal they shall show no cracks. Hydrogen embrittlement relief tests are not applicable to corrosion-resistant steel spring pins.

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4.8.9 Hardness.

4.8.9.1 Type I, slotted spring pins. Slotted spring pins shall be tested for hardness in accordance with MIL-STD-1312, test 6, hardness. The hardness shall be within the hardness limits of 3.6.

4.8.9.2 Type II, coiled spring pins. Coiled spring pins shall be tested for hardness on the inside surface of the outside half coil of the spring pin in accordance with MIL-STD-1312, test 6, hardness. The hardness shall be within the hardness limits of 3.6.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2).

6. NOTES

6.1 Intended use. Spring pins covered by this specification are intended to be used primarily for insertion in matching holes in adjoining pieces to position and hold these pieces in place.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and applicable specification sheet.
b. Applicable specification sheet part number(s).
c. Applicable levels of packaging (see 5.1).

Custodians:
Army - AR
Navy - OS
Air Force - 99

Review activities:
Army - ER, MI
DLA - IS
NSA-NS

User activities:
Army - AV, CE, ME
Navy - MC, YD

Preparing activity:
Army - AR

Agent:
DLA - IS

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